4 **(a)** Define *electric field strength*.

Electric field strength at a point in the electric field is the force per unit positive charge acting on a test charge at that point.

[1]

(b) Two charged metal spheres A and B are situated in a vacuum, as illustrated in Fig. 4.1.

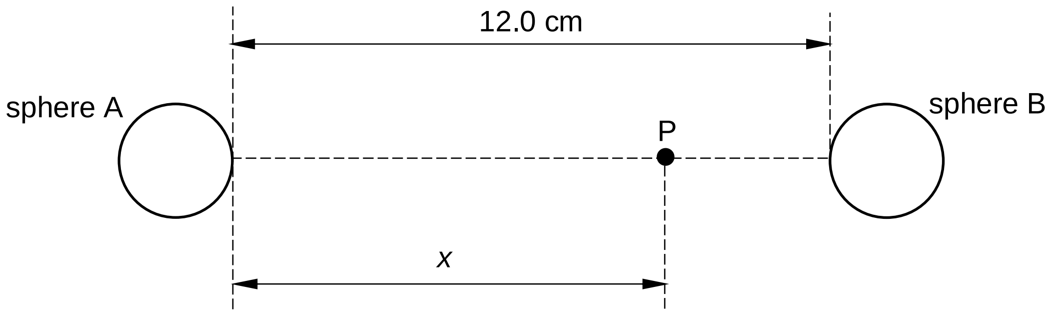

Fig. 4.1

The shortest distance between the surfaces of the spheres is 12.0 cm.

A movable point P lies along the line joining the centres of the two spheres, a distance x from the surface of sphere A.

The variation with distance x of the electric field strength E at point P is shown in Fig. 4.2.

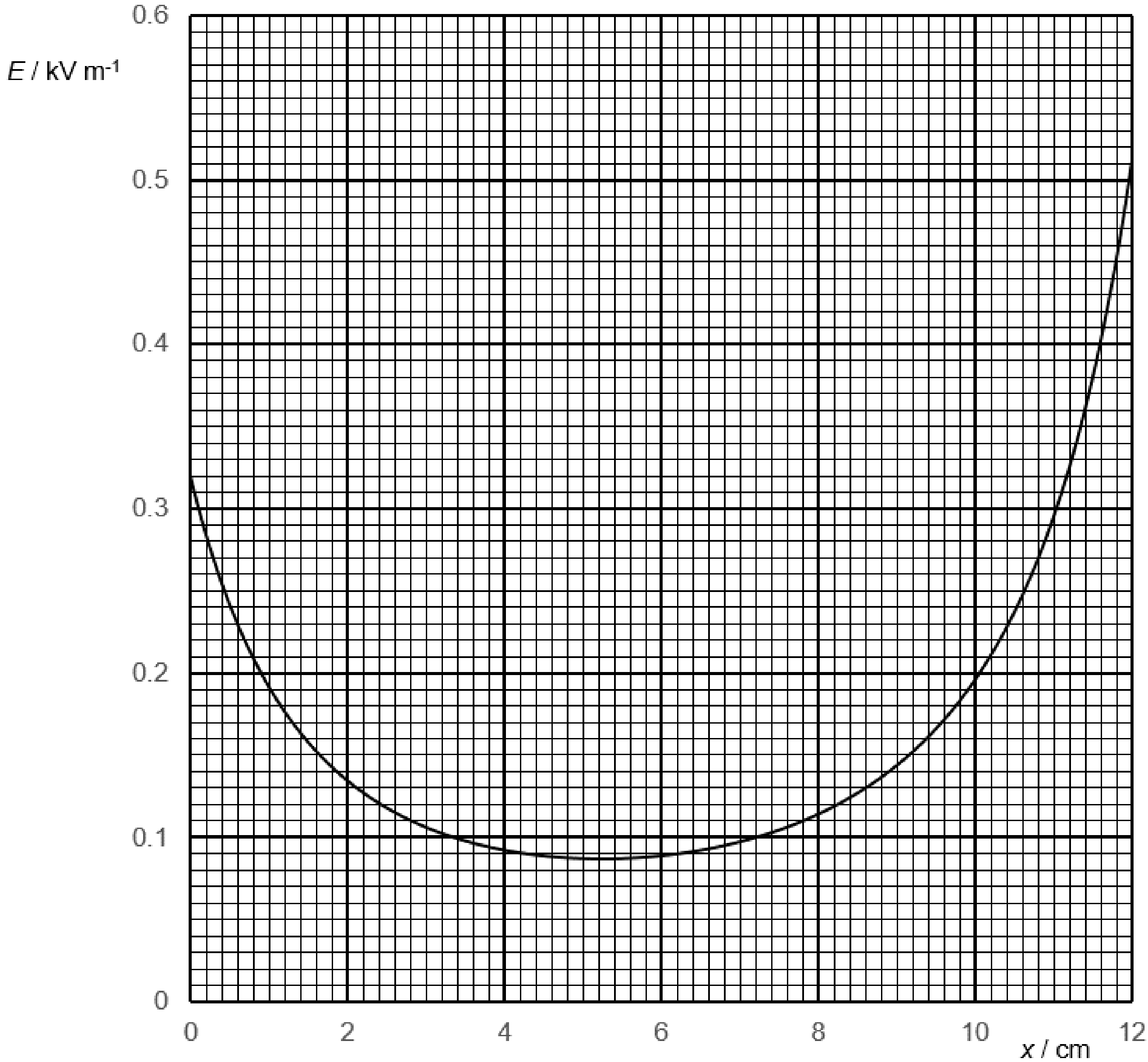

Fig. 4.2

(i) Using Fig. 4.2, comment on the magnitude and polarity of the charge of sphere A and sphere B.

Since the *E*-*x* graph is always positive, it shows that the field strength is acting in the same direction (to the right), therefore sphere A (positive) and sphere B (negative) must have different polarity. Sphere B has a larger magnitude of charge than sphere A. [2]

(ii) A proton is at point P where its acceleration is at a minimum. Use data from Fig. 4.2 to determine the acceleration of the proton.

From graph, minimum *E* is 0.088 kV.

By Newton's 2nd Law of motion,

$\Sigma F = ma \quad \rightarrow \quad eE = ma$

$a = \dfrac{eE}{m} = \dfrac{1.60\times10^{-19}(0.088\times10^{3})}{1.67\times10^{-27}} = 8.4\times10^{9}\ \text{m s}^{-2}$

acceleration = ... m s^{-2} [2]

(c) The variation with distance *x* of the electric potential V at point P is shown in Fig. 4.3.

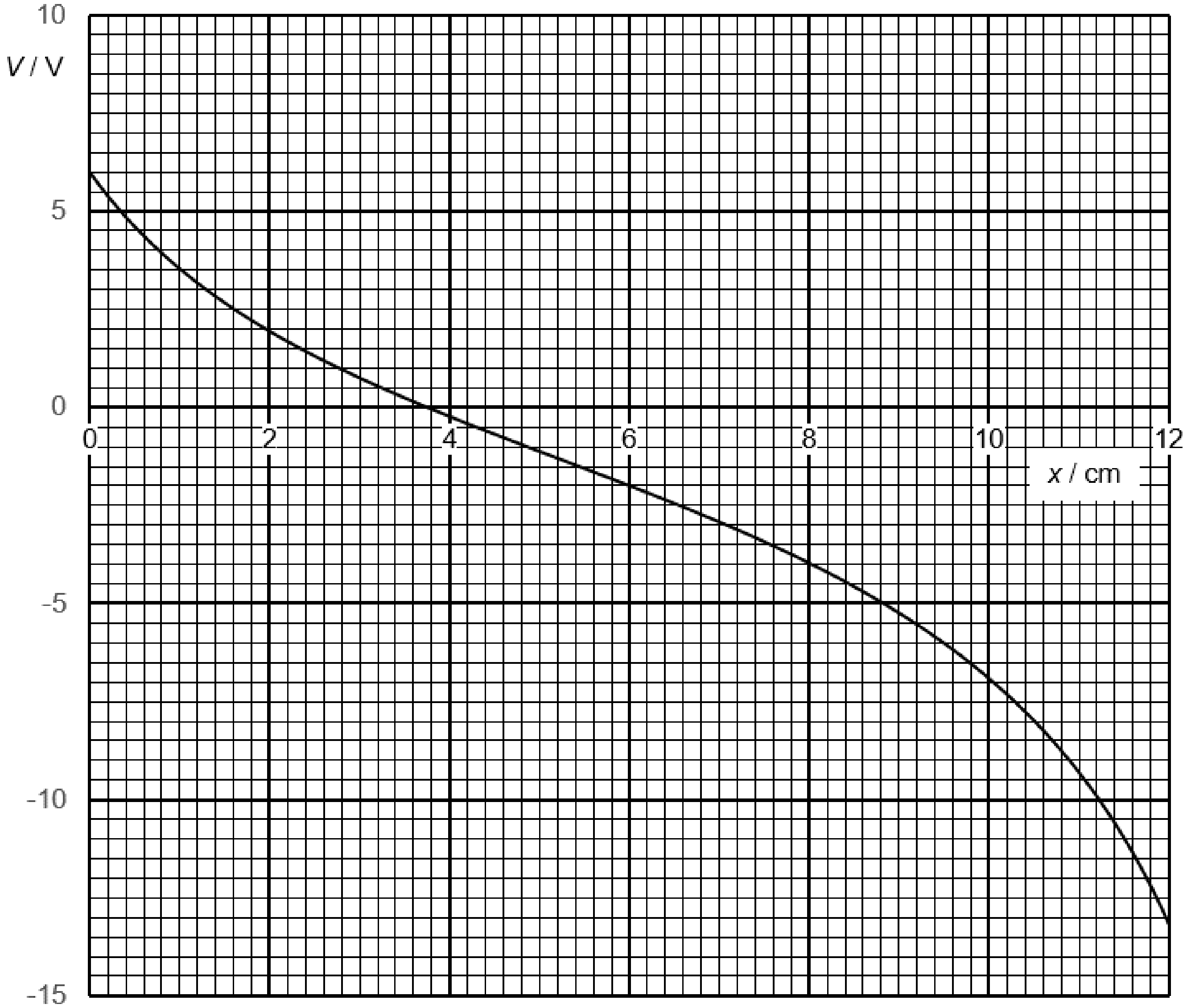

Fig. 4.3

(i) State, in words, the relationship between electric potential at point P and electric field strength at point P.

Electric field strength at point P is the negative potential gradient at point P.

....[1]

(ii) A proton is initially at rest on the surface of sphere A and moves along the line joining the centres of the two spheres to reach maximum speed at point P.

Use data from Fig. 4.3 to determine the maximum speed of the proton.

From graph, largest ΔV is – 13.2 – 6.0 = – 19.2 V.

By Conservation of Energy, Gain in KE = Loss in Elec PE

$$\frac{1}{2}mv_f^{\,2} - 0 = e\Delta V$$

$$v_f = \sqrt{\frac{2e\Delta V}{m}} = \sqrt{\frac{2(1.60\times10^{-19})(19.2)}{1.67\times10^{-27}}} = 6.07\times10^{4}\ \text{m s}^{-1}$$

maximum speed = ... m s^{-1} [3]

[Total: 9]

5 A cell P, a fixed resistor *R* and a uniform resistance wire AB are connected in a circuit as shown